Smart Library Management System

Final design revision · CPSC 315 · Michael Pelitera · Module 9

The Module 2 design document is kept unchanged as the historical record of what was designed. This page records what was actually built, and names every place the two differ.

Architecture as built

Presentation LibraryUI — console menus, input validation, role-based display

Coordination LibrarySystem — checkout, return, reserve, undo, reports, authorization

Structures BookInventory · UserAccounts · DueDateHeap · Recommender · ActivityLog · ReservationQueue

Records Book · User · Loan · ActivityEntry

Services PasswordHash over Sha256 · Huffman

Change from Module 2. That document showed four managers (Catalog, Account, Loan, Recommender) between the interface and the structures. They were not built. One coordination class, `LibrarySystem`, owns every operation that touches more than one structure. The rule that replaced them is narrower and easier to check: no structure module includes another structure module, so every cross-module operation has to pass through one file.

Structures, designed against built

Component	Module 2 design	As built
Book inventory	Hash table on ISBN	As designed, FNV-1a with chaining. Plus a secondary word index on titles and authors
User accounts	Linked list	Sorted doubly linked list plus a hash index from user ID to node
Activity log	Stack	As designed. Linked stack per account, driving undo
Reservations	Queue	As designed. Linked FIFO per title with a tail pointer
Due dates	Min-heap / priority queue	As designed, plus a loan ID to index map so early returns are O(log n)
Recommendations	Binary search tree, graph	As designed. Adjacency-list graph capped at 50 edges per title; score BST rebuilt per request
Passwords	Salted hash	PBKDF2-HMAC-SHA256, 16-byte salt, 600,000 iterations
`Book.activeHolds`	Per-title hold counter	**Deliberately not built.** The queue stores its own length in O(1); a second copy is the shape of both Module 6 defects
Fines, renewals	Borrowing rules	**Not built.** Recorded in the technical report

Screens as built

Patron	Staff adds
1. Search and borrow 2. Borrow by ISBN 3. Return a book 4. My account (loans with individual due dates) 5. Recommendations 6. Reserve a title that is out 7. My recent activity, with undo 8. Sign out	8. Overdue report 9. Inventory report 10. Hold lists 11. Storage report (compression) 12. Manage catalog and accounts 13. Sign out

Change from Module 2. The mock-ups showed `[H] place hold`, `[E] renew` and a fines line on the account screen. Holds were built and are option 6. Renewals and fines were not, so those options are absent rather than present and refusing. The account screen gained per-loan due dates, which the mock-up specified and the first build omitted.

Authorization

Role checks run in `LibrarySystem`, not in the interface. The console passes the authenticated account ID to each reporting and management method, which checks the role there and returns `NOT_AUTHORIZED`. Hiding staff options from a patron menu is convenience, not the control.

> **Scope, stated plainly.** The low-level structure accessors remain public for test fixtures and internal use, so the project does not claim that every public method enforces authorization. That is a limitation of a teaching implementation and is recorded in technical report section 9.

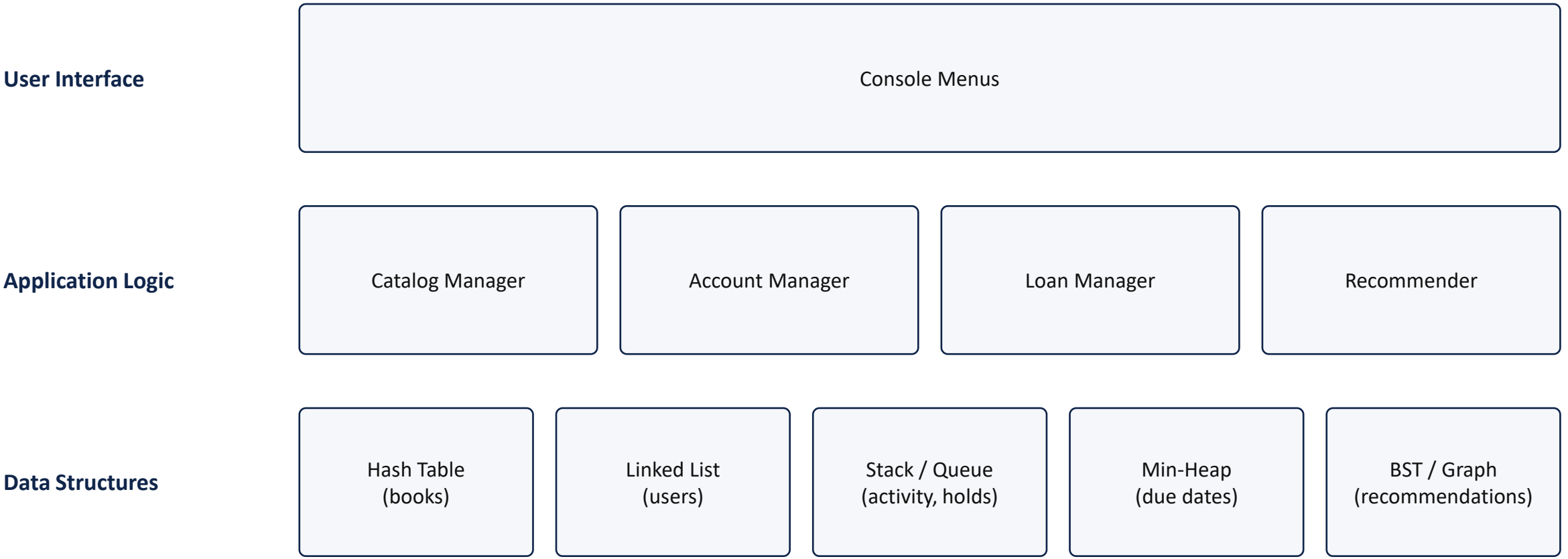
Sign-in

Account ID and password. Until Module 9 the screen took the ID alone: the `passwordHash` field existed and nothing read it.

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System Architecture



The console interface passes requests to four managers, and each manager owns the data structures for its part of the system.

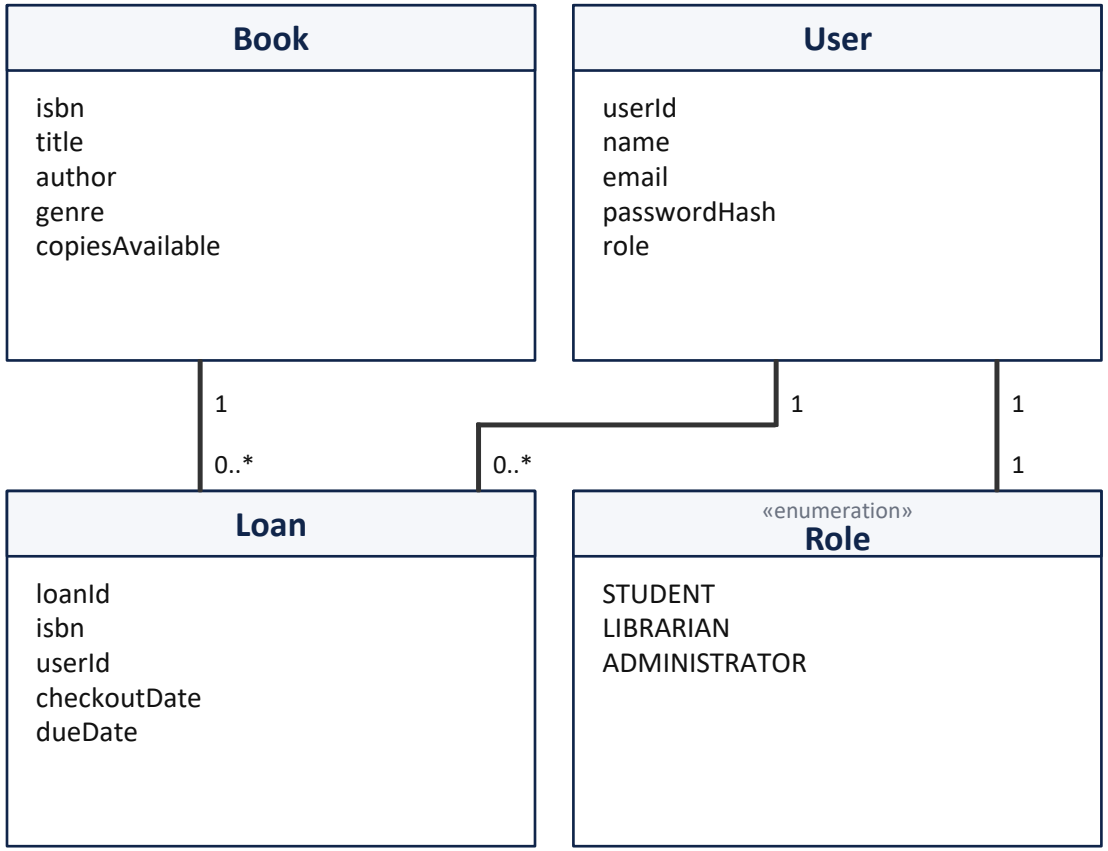
Data structures and class design

Each structure chosen for the operation it performs best.

Component	Structure
Book inventory	Hash table on ISBN
User accounts	Linked list
Activity log	Stack
Reservations	Queue
Due dates	Min-heap / priority queue
Recommendations	Binary search tree, graph
Passwords	Salted hash

Search approach

A hash table gives O(1) average lookup by ISBN. Binary search at O(log N) requires sorted data, so sorted lists by title or author are generated when a user requests them.



UML class diagram. Each Book relates to many Loans, and each User holds many Loans. Role determines which menu options a user sees.

Interface mock-ups

Menu-driven console. Every screen shows the signed-in user and the available actions.

Catalog search

```
+---------------------------------------------+
| SEARCH CATALOG              [Mike / Student]|
+---------------------------------------------+
| Search by: [1] Title  [2] Author  [3] ISBN  |
|                                             |
| Query: apollo_                              |
|                                             |
| 1. Digital Apollo        Mindell  2008  [2] |
| 2. Apollo Guidance Comp. O'Brien  2010  [0] |
|                                             |
| [#] borrow   [H] hold   [Q] back            |
+---------------------------------------------+
```

Student account

```
+---------------------------------------------+
| MY ACCOUNT                  [Mike / Student]|
+---------------------------------------------+
| Active loans: 2 of 5                        |
|                                             |
| Digital Apollo          due Sep 10          |
| The Right Stuff         due Sep 02  OVERDUE |
|                                             |
| Fines owed: $1.00                           |
|                                             |
| [R] return   [E] renew   [Q] back           |
+---------------------------------------------+
```

Borrowing rules: Five active loans per user, 21-day loan period, no checkout with unpaid fines, and no renewal when another user is waiting. Each rejection returns a specific reason, so the user knows which rule stopped them. Librarian and administrator screens use the same layout with additional options.